

# Road localization, then anomaly detection

UAV-based highway monitoring requires two sequential capabilities: localizing the road surface (**Stage 1**), then detecting anomalous objects within it (**Stage 2**). All candidates are evaluated **zero-shot** (no training or fine-tuning).

RECOMMENDATION

**Stage 1 (road localization):** Grounding DINO, road-focused prompt -- highest recall of the candidates, with a fallback required for the frames it fails to detect on at all. **Stage 2 (anomaly detection):** self-supervised feature matching with prototype distance. Closed-set segmentation with rejection is a viable Stage 2 alternative; the same language-prompt approach that leads Stage 1 is not recommended for Stage 2, where the task -- identifying a specific object rather than a broad surface -- does not favour it (see Stage 2 evidence).

|                      |                   |                  |               |
|----------------------|-------------------|------------------|---------------|
| PIPELINE STAGES      | STAGE 2 TEST BEDS | CANDIDATE MODELS | IMAGES SCORED |
| 2                    | 4                 | 8                | 940           |
| STAGE 2 "CAUGHT" BAR |                   |                  |               |
| own-frame top 10%    |                   |                  |               |

# Which model finds the highway first

No UAV highway dataset with road-region labels exists. AeroScapes (drone imagery, 5-50 m altitude, a labelled "road" class, though its scenes are parks and streets rather than highways) is used as a quantitative proxy; real UAVDT highway frames are checked qualitatively, against no ground truth.

**IoU:** overlap between predicted and true road area (1 = exact match, 0 = none).  
**Precision:** fraction of the predicted area that is actually road. **Recall:** fraction of the true road that was predicted.

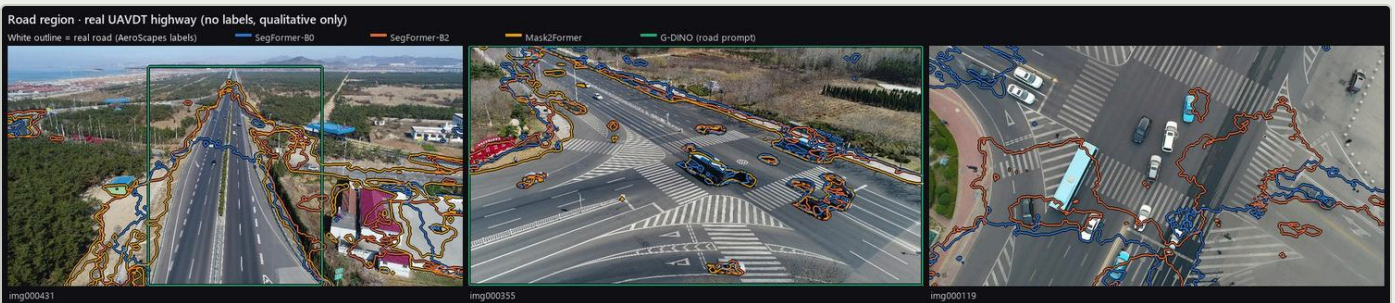
| MODEL                        | IOU                    | PRECISION |     | RECALL | SPEED / FRAME |
|------------------------------|------------------------|-----------|-----|--------|---------------|
| SegFormer-B0 (Cityscapes)    | <div><div></div></div> | 23%       | 46% | 35%    | 0.03s         |
| SegFormer-B2 (Cityscapes)    | <div><div></div></div> | 31%       | 52% | 53%    | 0.05s         |
| Mask2Former (Cityscapes)     | <div><div></div></div> | 13%       | 37% | 18%    | 0.10s         |
| Grounding DINO (road prompt) | <div><div></div></div> | 32%       | 37% | 79%    | 0.83s         |

**The closed-set segmenters fail for a specific, verified reason: viewpoint, not image resolution.** Per-pixel inspection shows SegFormer-B2 correctly labels 55% of true road pixels as "road"; the largest source of error is misclassification as "building" (29% of all true road pixels) -- a systematic misclassification, not noisy or low-confidence output. The error is viewpoint-dependent: on frames with an oblique, ground-level-like composition (buildings visible at their normal orientation, a horizon), SegFormer correctly labels most of the true road; on near-nadir frames it frequently labels the same paved surface "building" instead. Forcing the model's native 1024×1024 input (its processor defaults to a squashed, non-aspect-preserving 512×512) changes individual frames' scores in both directions but does not raise the aggregate mean -- ruling out preprocessing as the primary cause.



Labelled road (white) vs. predicted region. Grounding DINO (green) over-covers the road; the segmenters (blue/red/orange) under-cover it, consistent with the class-confusion finding above.

**For a region-narrowing filter, recall is the relevant metric:** a true road pixel excluded from Stage 1's region can never be examined by Stage 2, while an included non-road pixel only costs Stage 2 additional search area. Grounding DINO's recall (79%) substantially exceeds the segmenters' (53% SegFormer-B2, 18% Mask2Former), consistent with the same viewpoint robustness: open-vocabulary matching is not constrained to Cityscapes' closed-set class boundaries.



Real UAVDT highway frames, no ground truth -- qualitative only.

**Recommendation: Grounding DINO (road prompt), with a required fallback.** On an unselected sample of 40 real UAVDT highway frames, 10 (25%) returned zero detections -- a more severe failure than reduced precision, since Stage 2 then receives no candidate region at all. Deployment requires an explicit fallback (run Stage 2 on the full frame, or on SegFormer-B2's prediction) for such frames, rather than skipping them. Neither candidate is precise enough for hard cropping (see Limits).

## 4 test beds, one headline table

**Caught:** an object's peak score falls within the top 10% of its own image's background score distribution (no threshold shared across images or methods). **AUROC:** probability that a randomly chosen anomalous pixel scores above a randomly chosen normal pixel (0.5 = chance, 1.0 = perfect separation). Neither metric quantifies false-alarm rate (see Limits).

No public dataset combines UAV viewpoint, highway scene, and small real debris; each available dataset satisfies exactly two of the three. Evaluation is therefore split across 4 test beds, each holding two conditions fixed while varying the third, to isolate which factor drives a given score difference.

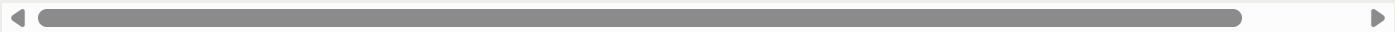
|           | TEST BED                                                                                               | IMAGES | LABELLED OBJECTS | OBJECT SHARE OF FRAME | SOURCE                    |
|-----------|--------------------------------------------------------------------------------------------------------|--------|------------------|-----------------------|---------------------------|
| <b>B1</b> | <b>Ground · road · salient anomalies</b><br>Animals, cones, boxes -- visually obvious                  | 70     | 253              | 0.16%                 | RoadAnc<br>EPFL + Anomaly |
| <b>B2</b> | <b>Ground · road · small obstacles</b><br>Boxes and boards on the road; only the road region is scored | 30     | 45               | 0.15%                 | SMIYC<br>RoadObs          |
| <b>B3</b> | <b>Near-ground · runway · mid-size debris</b><br>Airport runway debris, 300×300                        | 120    | 120              | 4.40%                 | FOD-A                     |

|  | TEST BED        | IMAGES | LABELLED OBJECTS | OBJECT SHARE OF FRAME | SOURCE |
|--|-----------------|--------|------------------|-----------------------|--------|
|  | close-up frames |        |                  |                       |        |

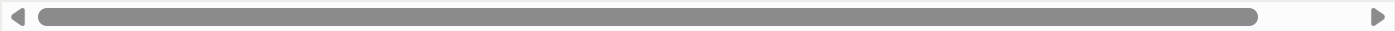
|    |                                                                                             |     |     |       |         |
|----|---------------------------------------------------------------------------------------------|-----|-----|-------|---------|
| B4 | Aerial · rail corridor · mid-size debris<br>Rail-corridor obstacles filmed at 10 m altitude | 120 | 120 | 3.67% | UAV-RSC |
|----|---------------------------------------------------------------------------------------------|-----|-----|-------|---------|

All models emit raw, unnormalised scores. An object is scored only against its own image; no threshold is shared across test beds or methods.

| MODEL                                               | B1                                 | B2                                  | B3                                  | B4                                       |
|-----------------------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|------------------------------------------|
|                                                     | Ground · road · salient anomalies  | Ground · road · small obstacles     | Ground · runway · medium obstacles  | Aerial · rail corridor · mid-size debris |
| Language-prompt detection<br>Grounding DINO Tiny    | hit <div></div> 48%<br>AUROC 0.854 | hit <div></div> 80%<br>AUROC 0.932  | hit <div></div> 57%<br>AUROC 0.781  | hit <div></div> 3%<br>AUROC 0.700        |
| Self-supervised features<br>DINOv2 ViT-L/14         | hit <div></div> 70%<br>AUROC 0.922 | hit <div></div> 100%<br>AUROC 0.992 | hit <div></div> 100%<br>AUROC 0.938 | hit <div></div> 100%<br>AUROC 0.892      |
| Self-supervised features<br>DINOv3 ViT-L/16         | hit <div></div> 57%<br>AUROC 0.889 | hit <div></div> 67%<br>AUROC 0.829  | hit <div></div> 100%<br>AUROC 0.908 | hit <div></div> 96%<br>AUROC 0.799       |
| Closed-set + rejection<br>SegFormer-B2 (Cityscapes) | hit <div></div> 74%<br>AUROC 0.804 | hit <div></div> 100%<br>AUROC 0.974 | hit <div></div> 99%<br>AUROC 0.752  | hit <div></div> 89%<br>AUROC 0.606       |



DINOv2 ranks first on every test bed and attains the highest AUROC on every test bed. Closed-set rejection ranks second but degrades most sharply in aerial views; language-prompt detection ranks last throughout. All four methods score lowest on B1, where the scored region is the full frame -- including sky, buildings, and vegetation -- rather than the road alone.



VERDICT BY METHOD FAMILY



**Recommended** **Self-supervised features + prototype distance**

DINOv2 ViT-L/14

Scores a region by similarity to the rest of its own frame, requiring no prior recognition of object identity or scene type. Ranks first on every test bed. A DINOv3 backbone was tested under the same protocol: 96% vs 100% recall on B4 -- no measurable advantage over DINOv2.

CAUGHT IN THE AIR (B4) **B4****100%**RANKING ABILITY (AUROC) **B4****0.892****Alternative** **Closed-set segmentation + rejection**

SegFormer-B2 (Cityscapes)

Comparable to the top-ranked method on ground-view test beds, but degrades most sharply in aerial views: its criterion depends on Cityscapes' 19 semantic classes, which do not hold under a nadir viewpoint.

CAUGHT IN THE AIR (B4) **B4****89%**RANKING ABILITY (AUROC) **B4****0.606****Not recommended** **Open-vocabulary detection via language prompts**

Grounding DINO Tiny

Requires object recognition prior to flagging; object appearance under a nadir viewpoint differs substantially from ground-level training data. Ranks last on every test bed, with the largest decline in aerial views.

CAUGHT IN THE AIR (B4) **B4****3%**RANKING ABILITY (AUROC) **B4****0.700**

STAGE 2 · ANALYSIS

## Two factors matter more than the model

Object scale relative to the feature grid, and camera viewpoint, account for more of the score variation than model choice.

OBJECT SCALE

Self-supervised methods partition each frame into a feature grid and compare individual cells to the frame mean. B2's median object spans **2.2 feature cells** at 896 px input resolution -- marginally more than one. Increasing input resolution alone (224→896 px, model and weights unchanged) raises B2 recall from 76% to 100% and AUROC from 0.836 to 0.992, at a cost of 0.10 GB additional VRAM (within the 4 GB budget). **Flight altitude, sensor resolution, and input resolution jointly determine more of the outcome than model selection.**

| INPUT SIZE | FEATURE GRID | PEAK VRAM | B2 CAUGHT (OWN-FRAME TOP 10%) | RANKING |
|------------|--------------|-----------|-------------------------------|---------|
| 224×224    | 16×16        | 0.58 GB   | <div><div></div></div> 76%    | 0.836   |
| 448×448    | 32×32        | 0.60 GB   | <div><div></div></div> 91%    | 0.966   |
| 672×672    | 48×48        | 0.64 GB   | <div><div></div></div> 98%    | 0.984   |
| 896×896    | 64×64        | 0.69 GB   | <div><div></div></div> 100%   | 0.992   |

VIEWPOINT

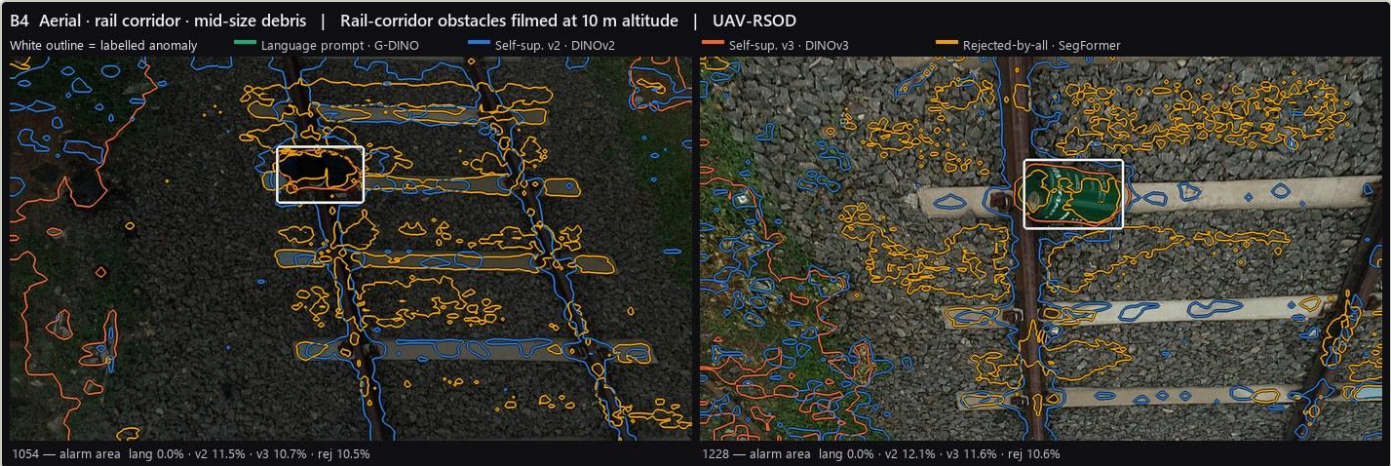
B3 (near-ground, runway) and B4 (UAV, 10 m, rail corridor) are matched on object scale and background material; camera elevation is the primary remaining difference. DINOv2 recall declines by 0 points, consistent with a criterion independent of object or scene identity. Rejection declines by 10 points, the largest drop, consistent with dependence on Cityscapes semantic classes that do not hold under a nadir viewpoint.

| MODEL                                          | B3 NEAR-GROUND | B4 AERIAL | CHANGE  |
|------------------------------------------------|----------------|-----------|---------|
| Language-prompt detection(Grounding DINO Tiny) | 57%            | → 3%      | -53 pts |
| Self-supervised features(DINOv2 ViT-L/14)      | 100%           | → 100%    | +0 pts  |

| MODEL                                     | B3 NEAR-GROUND | B4 AERIAL | CHANGE  |
|-------------------------------------------|----------------|-----------|---------|
| Self-supervised features(DINOV3 ViT-L/16) | 100%           | → 96%     | -4 pts  |
| Closed-set + rejection(SegFormer-B2)      | 99%            | → 89%     | -10 pts |



B2 · Small road obstacles. Once input size is fixed, self-supervised (blue) hugs them.White outline = labelled anomaly; colours are each method's alarm region (own-frame top 10%).



B4 · UAV rail corridor. Self-supervised hugs the object; rejection (yellow) scatters across the ballast; language-prompt (green) barely fires.White outline = labelled anomaly; colours are each method's alarm region (own-frame top 10%).

**Model-selection rationale:** the two-stage design pairs a high-recall region proposer (Grounding DINO, road-focused prompt, Stage 1 -- prioritising road coverage over precision, with a fallback for its zero-detection frames) with a viewpoint-robust, scene-agnostic anomaly detector (DINOv2 prototype matching, Stage 2). Each choice is

supported by measurement across multiple test beds rather than a single scene. The pipeline has not been validated end-to-end; the following limitations qualify both stages.

## What this does not prove, and what's next

Limitations that qualify the recommendation above.

- ❗ **No test bed matches the target domain, in either stage.** B4 (Stage 2) is a rail corridor, not a highway; AeroScapes (Stage 1) is parks and streets, not highway. Claims about highway performance are extrapolated from adjacent conditions, not measured directly.
- ❗ **Stage 1 output is a coarse region, not a precise mask.** Grounding DINO's precision is approximately 0.37: the predicted region is substantially larger than the true road. Sufficient to narrow attention away from clearly irrelevant background, not to hard-crop a frame down to road-only pixels without excluding adjacent context Stage 2 may need. It also returns zero detections on approximately one in four real highway frames sampled, requiring the fallback described in Stage 1.
- ❗ **Only the Stage 2 recommendation has a family-level result.** DINOv2 and DINOv3 agree; the other two families were each tested with one checkpoint, so those results describe that checkpoint specifically, though the failure mode is consistent with the family's known mechanism.
- ❗ **Box annotations inflate absolute scores** on UAV-RSOD and FOD-A (the full bounding box is scored as anomalous). This affects all four methods equally, so the ranking is unaffected, but absolute values are not comparable across annotation types.
- ❗ **The recommended Stage 2 method has a known failure mode:** once anomalies exceed approximately 40% of the road area, the method inverts, classifying clean road as anomalous. Highway debris is well below this threshold.
- ❗ **False-alarm rate is not measured, by design.** UAVDT is the only UAV-highway dataset available but has no road-region labels; scoring the full frame would count correct detections of buildings or billboards as false alarms solely for being off-road, and Stage 1 accuracy is insufficient to correct this reliably. This measurement is deferred until Stage 1 accuracy improves and a downstream reasoning stage can assess relevance.
- ➔ **Next steps:** improve Stage 1 accuracy (post-processing to reduce prediction fragmentation, or combining segmenter precision with the road-prompted detector's recall) to a level suitable for frame masking; reintroduce false-alarm measurement; and construct a synthetic evaluation set combining real highway footage with inserted debris, satisfying all three Stage 2 conditions simultaneously.

```
results/benchmark.json + results/segmentation_benchmark.json · Stage 2 "caught" bar: own-  
frame top 10% · zero-shot, no fine-tuning
```